

Product Requirements Document

Proposal to Close the Android 5.4 KYC Completion Gap — PulsePay Product Analytics

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1. Problem

Users on Android app version 5.4 completed KYC (Know Your Customer identity verification) at a materially lower rate than users on the prior stable version, Android 5.3. Of 5,150 KYC attempts analyzed, only 2,810 completed successfully (54.56%), a 21.04 percentage point gap versus the 75.60% completion rate on Android 5.3. The affected version also showed elevated document-upload errors (1,476 recorded), pointing to the document-upload step as the likely failure point.

2. Why it matters

Using Android 5.3 as a counterfactual baseline, Android 5.4 users would be expected to produce roughly 3,894 completions at parity performance — about 1,084 more than actually occurred. Given a 57.62% post-KYC activation rate, this translates to an estimated 624 potentially lost activated users. These are counterfactual estimates, not proven causal impact, but the size of the gap and the concentration of upload errors make this worth prioritizing.

3. Goal & success metric

- Primary goal: close the KYC completion-rate gap between Android 5.4 and the Android 5.3 baseline.
- Success metric: Android 5.4 KYC completion rate reaches parity (~75%) with Android 5.3 within one release cycle of the fix shipping.
- Guardrail metric: support contact rate during KYC does not increase, and downstream activation rate does not decline.

4. User story

As a user completing KYC on Android 5.4, I want the document-upload step to work reliably, so that I can finish identity verification without errors and activate my account.

5. Proposed scope

- In scope: root-cause investigation of document-upload failures specific to Android 5.4 (e.g., camera/file-picker permissions, image compression, upload timeout handling).
- In scope: version-level monitoring/alerting on KYC completion rate and upload error rate, so future regressions are caught before they reach this scale.
- Out of scope (this phase): broader KYC flow redesign; that is covered separately by the KYC A/B experiment (see Section 7).

6. Supporting data

Metric	Value
KYC completion rate — Android 5.4 (affected)	54.56%
KYC completion rate — Android 5.3 (baseline)	75.60%
Completion-rate gap	-21.04 pp
Est. additional completions at baseline rate	+1,084
Est. additional activated users	+624

7. Related experiment finding (context, not a fix)

A separate A/B test simplifying the KYC flow (6,032 users) showed a directional but not statistically significant improvement in completion rate (+1.55pp, p=0.22), and flagged a sample-ratio mismatch (SRM p=0.0268) suggesting an assignment/instrumentation issue. Recommendation: do not treat the simplified flow as a validated fix; investigate

assignment integrity before re-testing. This PRD's Android 5.4 fix is a separate, higher-confidence root cause and should be prioritized independently of the experiment result.

8. Open questions / risks

- Confirm whether the upload failure is device-side (OS API behavior) or app-side (a regression introduced in the 5.4 release).
- Validate the counterfactual assumption — would Android 5.4 users truly convert at the 5.3 baseline rate absent the bug?
- Resolve the experiment's sample-ratio mismatch before using experiment data to inform this fix's design.

9. Note on data source

This PRD is based on the PulsePay project — a real-world product analytics portfolio project investigating user activation, retention, acquisition quality, a KYC conversion incident, and an A/B experiment for a fictional fintech payments product. It does not represent a real company or real user data; it demonstrates the analysis-to-requirements workflow.